Jacob Forrette

Swansea MA, 5085253220, jacobforrette@gmail.com , https://www.linkedin.com/in/jacob-forrette/

EDUCATION

Wentworth Institute of Technology Boston, MA
Bachelor of Science in Electromechanical Engineering August 2026
Cumulative GPA: 3.75/4.0:
Relevant Coursework: Engineering Mechanics and Dynamics, Electrical Analysis, Analog and Digital Circuit Design, Advanced Math and Physics, Fluid Mechanics, Microcontrollers in C Programing, Thermodynamics, Engineering Design, Signals and System, Responsible Use of Generative AI, Feedback and Control, Heat Transfer, Electromechanical Systems

Wentworth Institute of Technology Boston, MA
Master of Science in Mechanical Engineering Expected August 2027
Cumulative GPA: 3.8/4.0
Relevant Coursework: Internal Combustion Engines, Reliability In Machine Design, Spatial Mechanism of Robotics, Design of Manufacturing,

SKILL AND CERTIFICATIONS

Software Skills: CAD Modeling MATLAB, TensorFlow, Machine Learning, Data Analysis, Arduino IDE, Multisim, Quartus, FPGA Design, Embedded Programming, Microsoft Office, Google Workspace
Technical Skills:, Soldering, Machining, Metal Casting, CNC Operations, CNC Programming, CNC Setup and Calibration, CNC Troubleshooting, Technical Drawing Interpretation, GD&T Interpretation, Strength and Load Testing, Function Generator, Digital Multimeter, Oscilloscope, Micrometers, Calipers, Microcontrollers, Texas Instruments MSP430, FPGA (Altera Cyclone II), Circuit Debugging, Assembly and Fit-Up, Quality Inspection,
Certifications & Training: CPR and First Aid Trained, Forklift Operation, CNC Programming, CMMC, Smart Manufacturing.
Awards: Dean's List 2022-2026, Honors Program

WORK EXPERIENCE

DW Clark Taunton, MA
Mechanical Engineering CO-OP Full Time Dec 2024- May 2025
Part Time May 2025- Aug 2025

- Supported the full casting lifecycle, including design, mold modeling, melting and pouring operations, post-processing, and non-destructive testing of cast components.
- Operated and maintained CNC machining equipment to manufacture parts to specification from engineering drawings and GD&T requirements
- Assisted with CAD modeling and collaborated with engineering staff to refine casting designs and optimize downstream machining processes.

DW Clark Taunton, MA
Production and Operation CO-OP Aug 2025- Dec 2025

- Troubleshot and improved mechanical systems, electrical components, and CNC controller logic across large CNC machines, material handling manipulators, and heat treat work centers
- Led startup, commissioning, and overhaul of multi-axis CNC vertical lathes, coordinating mechanical repairs, controller updates, and external service providers.
- Designed and implemented machining methods, workholding, and ring segmenting processes to improve efficiency and repeatability.
- Developed and deployed preventative maintenance procedures and MES workflows, enabling digital tracking and improved equipment uptime.
- Designed improved in-process material staging and inventory systems to reduce downtime and support daily production.

UNIVERSITY PROJECTS

Automated Safety Inspection System (ASIS) – Senior Design Jan 2025 – Aug 2026
Mechanical Design & Sensor Integration Lead

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- Designed and developed a remotely operated robotic inspection system to monitor hazardous environmental conditions in confined spaces using integrated gas, temperature, humidity, and proximity sensors.
- Integrated MQ-series gas sensors, oxygen (O_2), carbon dioxide (CO_2), BME280 environmental sensors, LiDAR, and a camera system with Arduino and Raspberry Pi controllers for environmental monitoring and obstacle detection.
- Designed and manufactured a custom steel chassis, sensor mounting components, ventilation system, and electrical enclosures using CAD modeling, 3D printing, and fabrication techniques.
- Developed an 18V battery-powered electrical system incorporating voltage regulators, circuit protection, cooling fans, and power distribution for sensors, motors, and controllers.
- Performed structural FEA, thermal analysis, sensor calibration, electrical troubleshooting, and functional testing to evaluate system reliability and performance in confined environments.

Cast in Steel Competition 2026 Aug 2025-Apr 2026
Horsemans Axe Team Lead

- Directed a five-member engineering team through design, analysis, casting, and validation of a competition-grade Horseman's Axe in compliance with SFSA requirements.
- Oversaw CAD modeling, mass distribution, and geometry refinement to meet performance, durability, and weight constraints while maintaining manufacturability.
- Planned and coordinated casting methodology, including mold design considerations, material selection, and post-cast processing to achieve structural integrity and surface quality.
- Managed project scheduling, technical documentation, and verification artifacts, ensuring completion of the formal technical report and presentation video to SFSA standards.
- Served as technical liaison between faculty advisors, foundry partners, and machinists, aligning design intent with real-world manufacturing and testing constraints.

ACTIVITIES

Boston College Marching Band and Pep Band Boston, MA
Tuba player Aug 2022- Present
Section Leader Nov 2022 – Present
American Foundry Society Boston, MA
Member Nov 2023- Present
SFSA Cast in Steel Boston, MA
Member Nov 2023 - Present
Team Lead Aug 2024 - Present